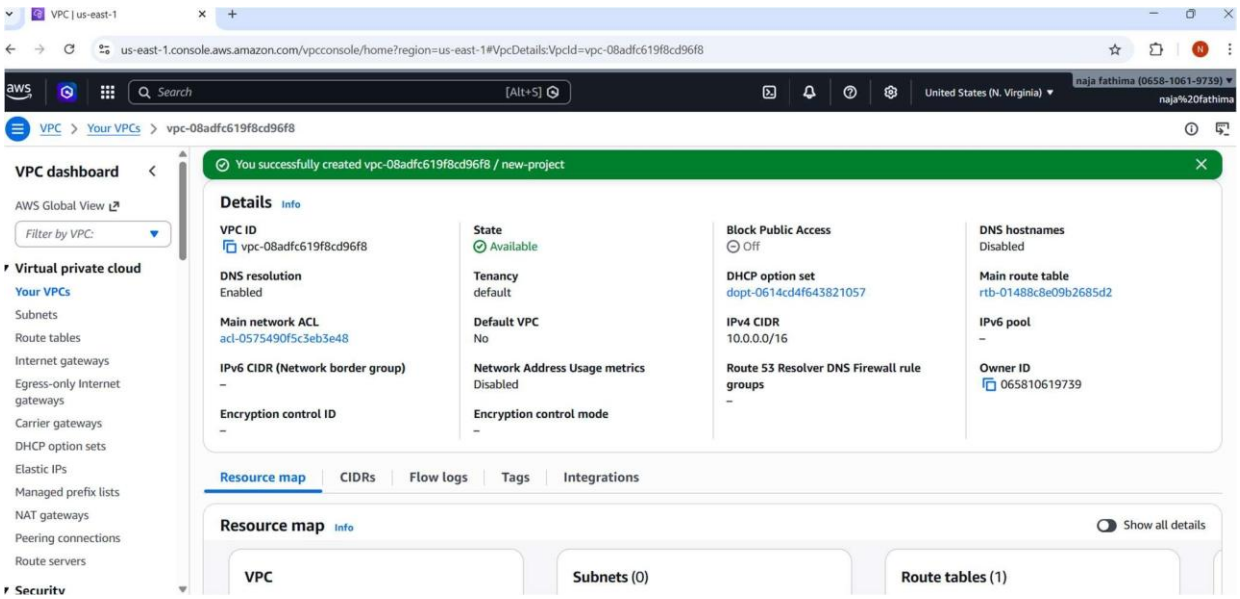
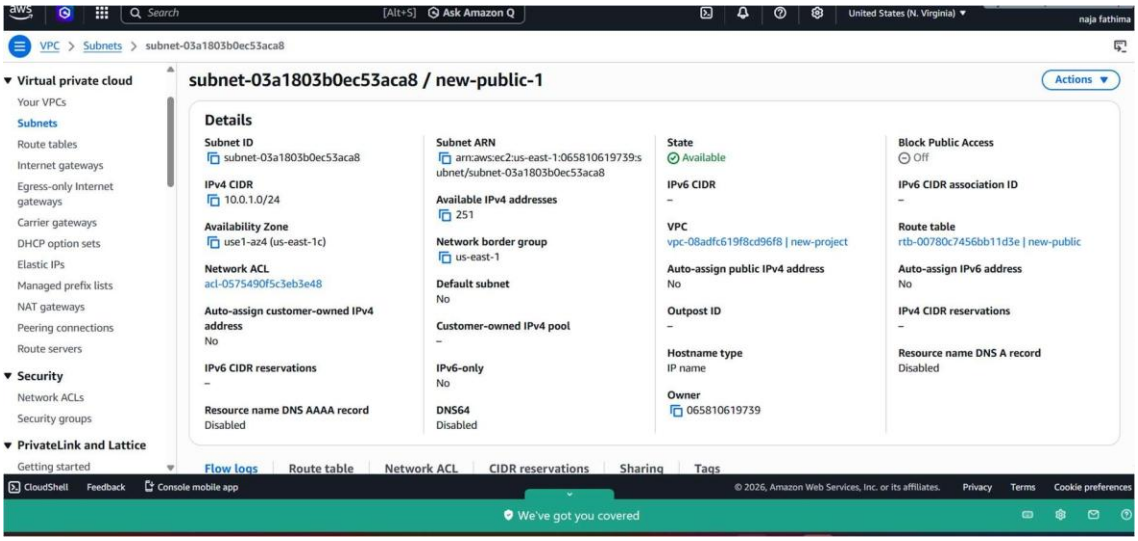
Step1: Created a VPC with CIDR 10.0.0.0/16. Created two public and private subnets. A ached Internet Gateway to the VPC.

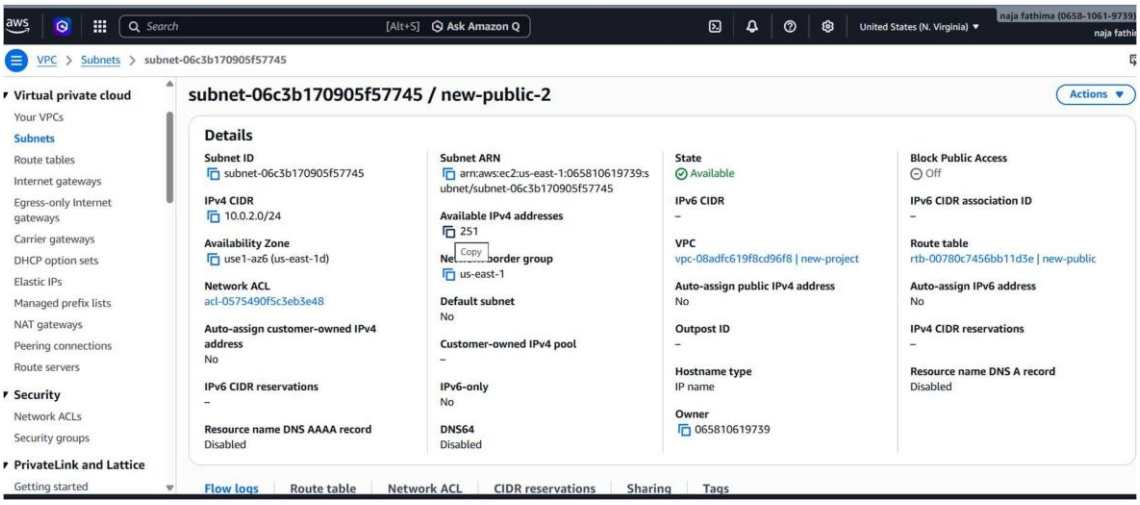
Created a VPC named new-project with CIDR 10.0.0.0/16.



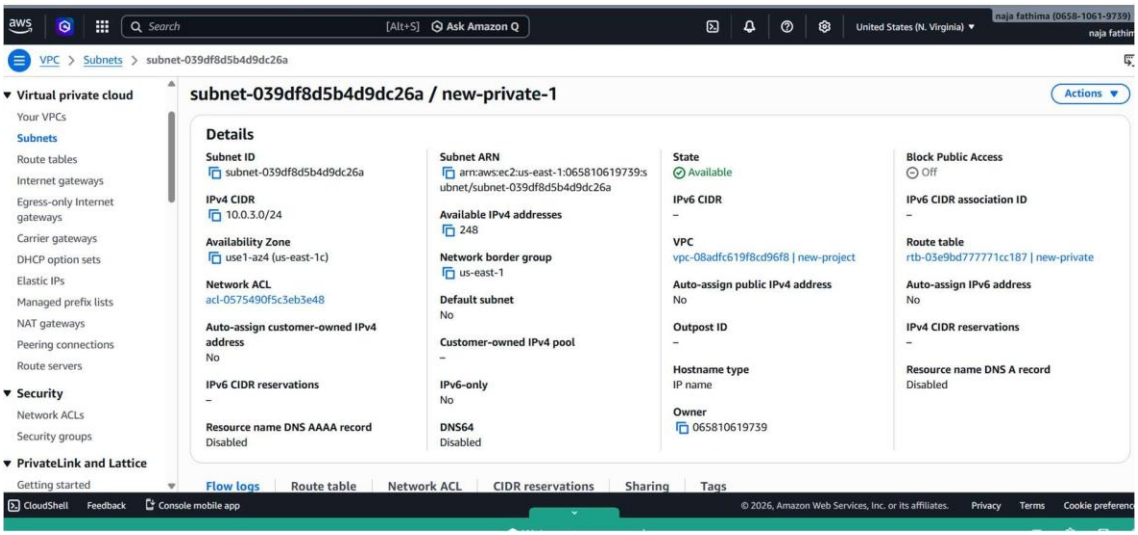
Created a public subnet named new-public-1 with subnet CIDR 10.0.1.0/24 in us-east-1c availability Zone.



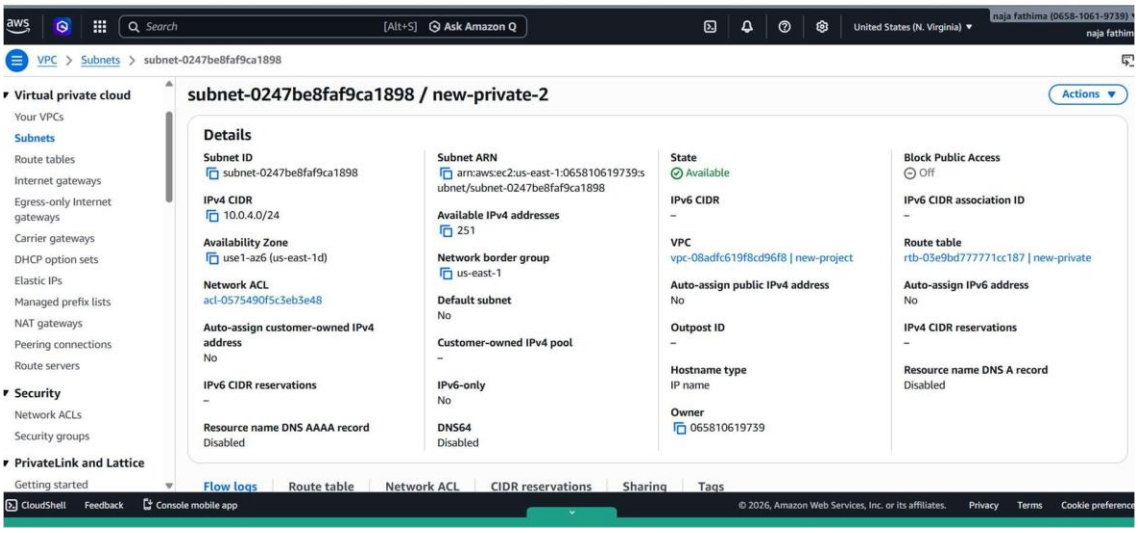
- Created another public subnet named newpublic-2 with subnet CIDR 10.0.2.0/24 in useast-1d availability Zone.


aws
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
naja fathima (0658-1061-9739)
naja fathi
VPC > Subnets > subnet-06c3b170905f57745
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
subnet-06c3b170905f57745 / new-public-2
Actions
Details
Subnet ID
subnet-06c3b170905f57745
IPv4 CIDR
10.0.2.0/24
Availability Zone
use1-az6 (us-east-1d)
Network ACL
acl-0575490f5c3eb3e48
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:us-east-1:065810619739:subnet/subnet-06c3b170905f57745
Available IPv4 addresses
251
Copy
us-east-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-08adfc619f8cd96f8 | new-project
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
065810619739
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-00780c7456bb11d3e | new-public
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags

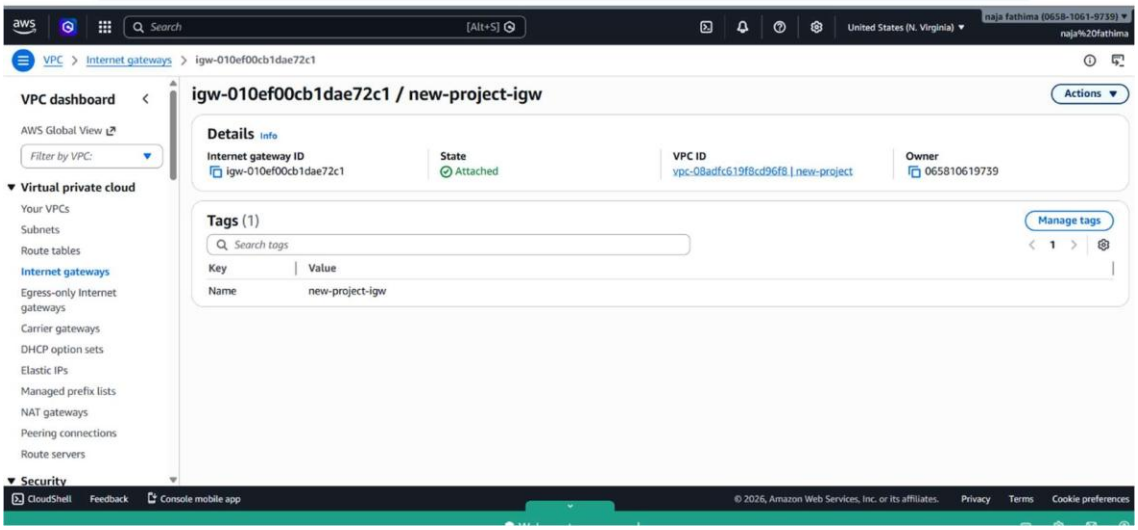
Created a private subnet named new-private1 with subnet CIDR 10.0.3.0/24 in us-east-1c availability Zone.



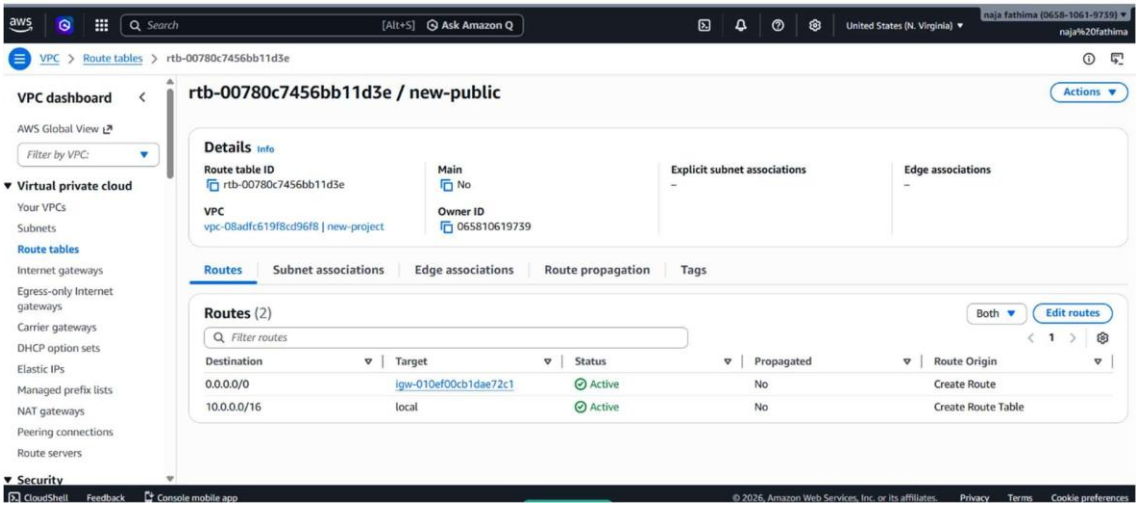
- Created another private subnet named newprivate-2 with subnet CIDR 10.0.4.0/24 in useast-1d.



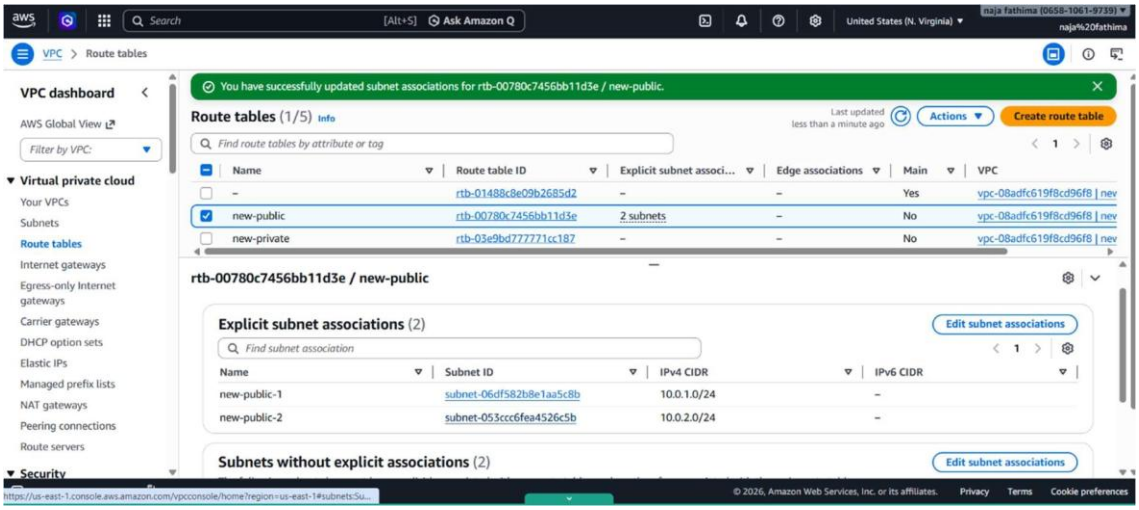
Created an Internet Gateway named new-project-igw and a ached to the VPC.



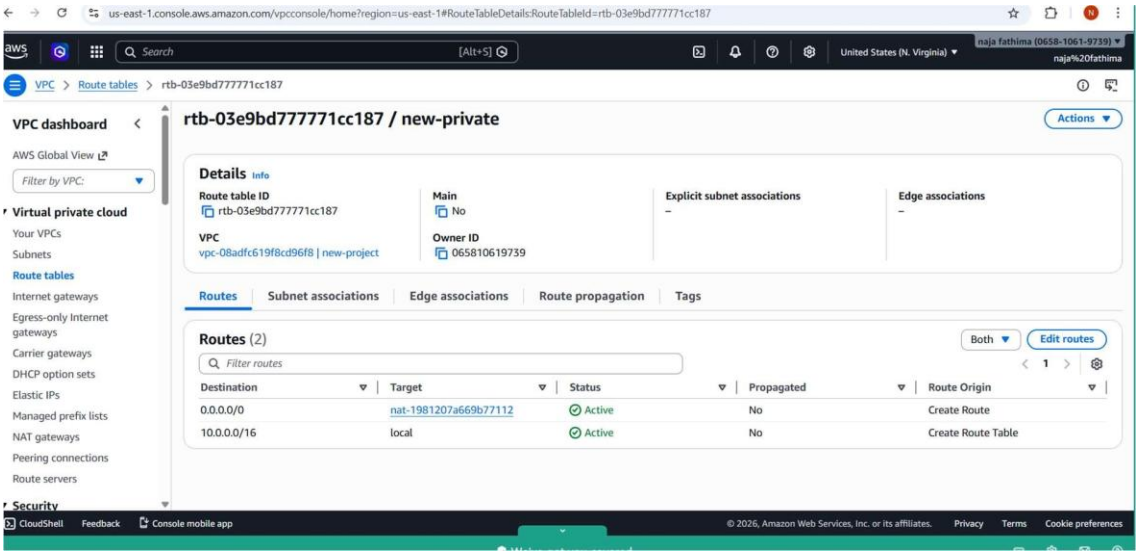
- Created a route table named new-public for public subnets which enable the internet access to the public subnets.



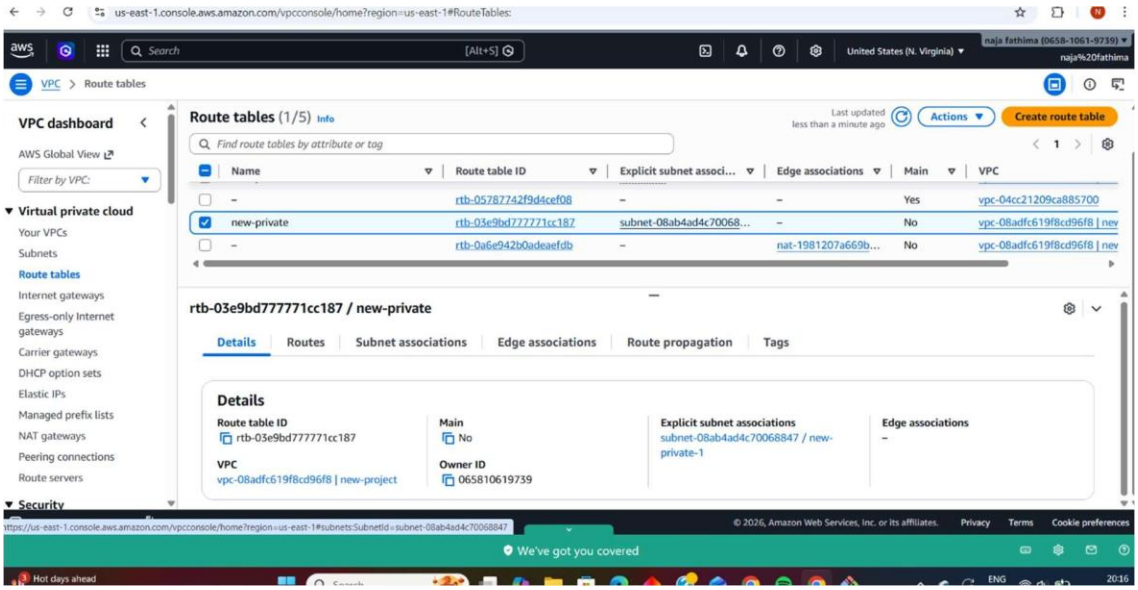
- Added two public subnets to the public route table.



- Created a route table named new-private for private subnet and configured NAT gateway for EC2 to install h pd package.

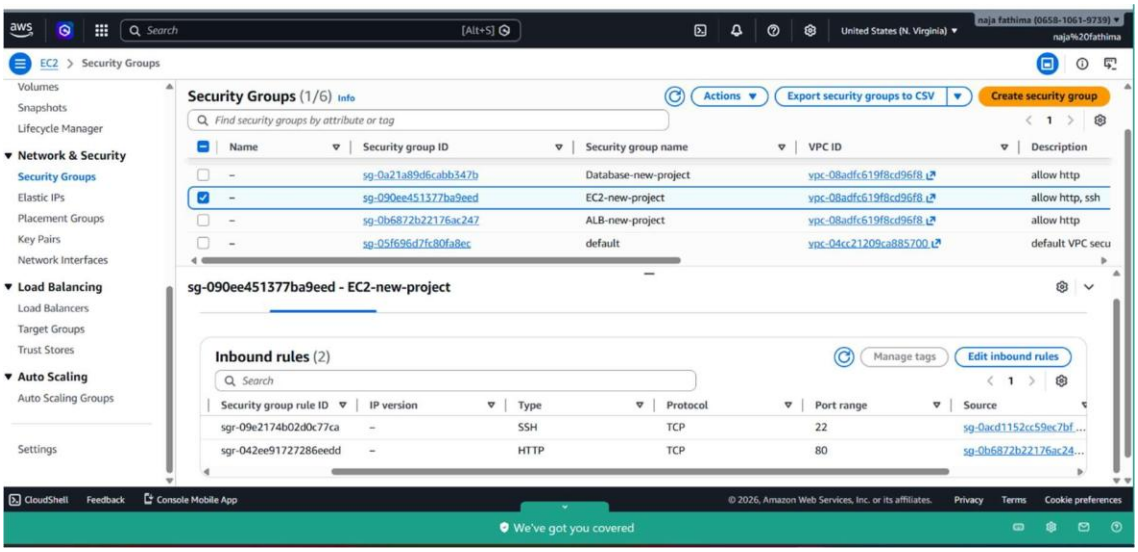


- Added new-private-1 subnet to the private route table.

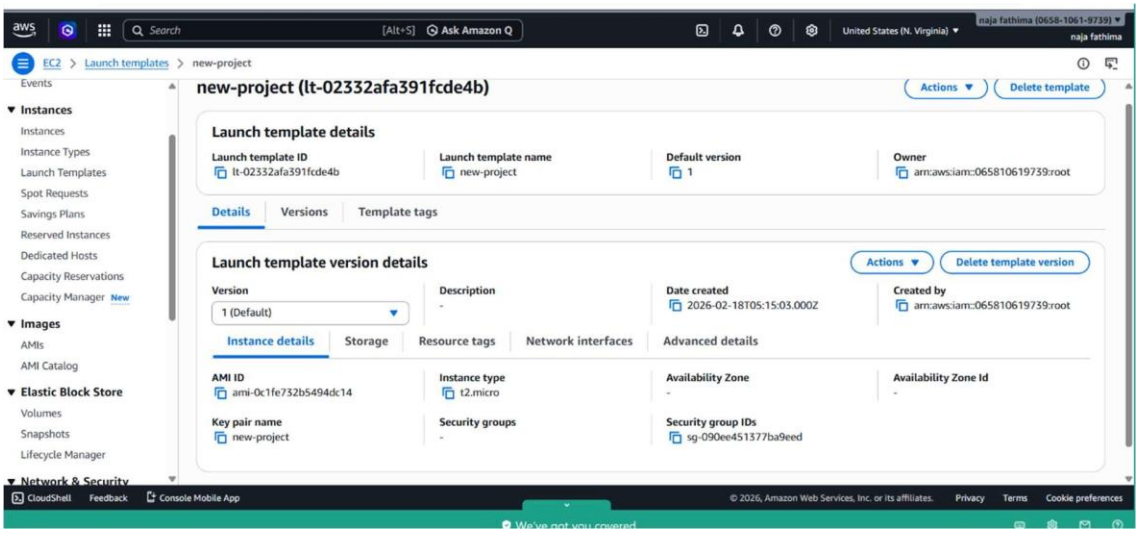


Step 2: Launched an EC2 Instance using launch template & User data.

- Created a security group for EC2 Instance which allow only HTTP request from ALB and SSH from Bas on host.



Created a launch template named newproject.



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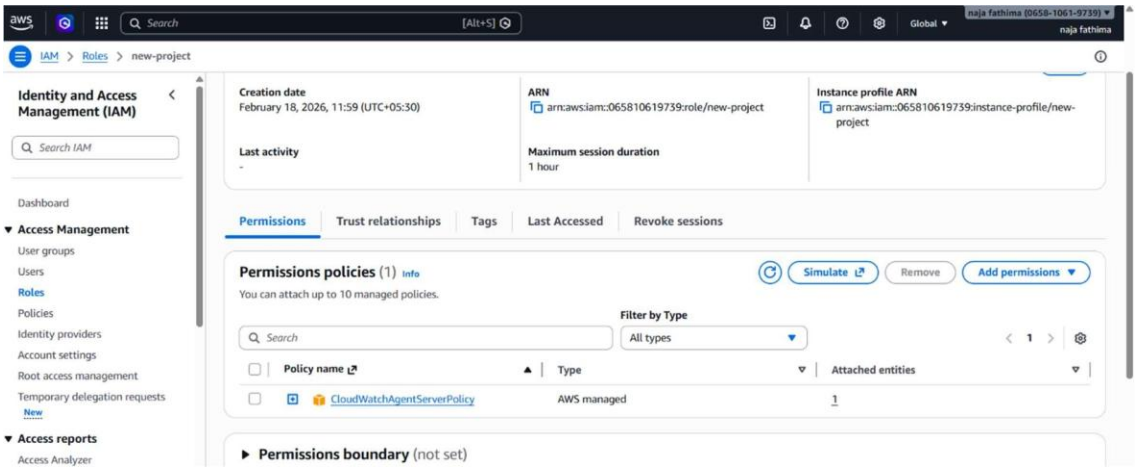
- Created an IAM role with CloudWatch permission and attached to the launch template.



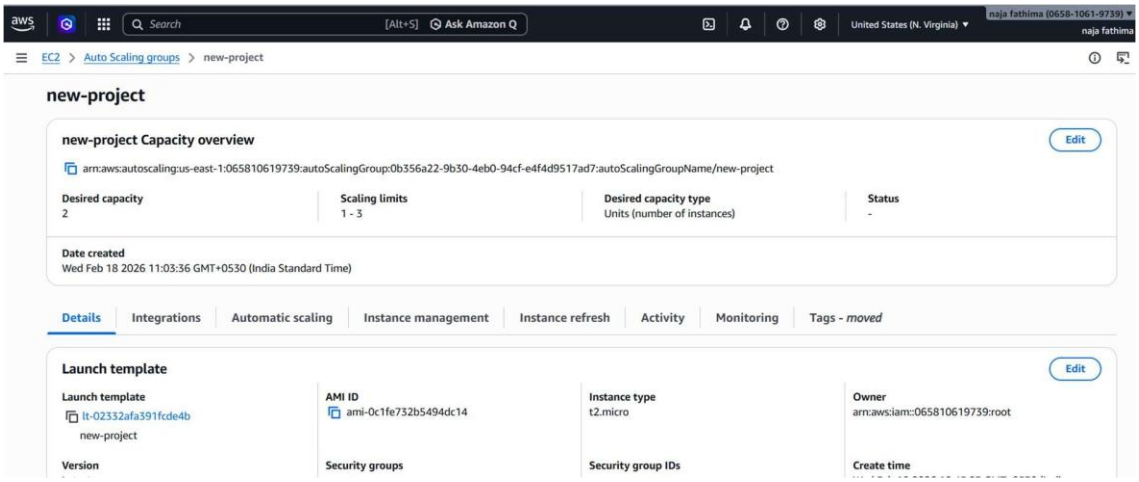
- Launched an EC2 Instance using launch template in new-private-1 subnet.
- Created a bastion host in public subnet which allow only SSH from my IP.
- Using bastion host SSH into the EC2 Instance in private subnet.

- Checked the status of httpd .
- Checked the application locally using curl localhost.

```
NAJA@LAPTOP-D9BDKEQK MINGW64 ~
$ cd Downloads/

NAJA@LAPTOP-D9BDKEQK MINGW64 ~/Downloads
$ ssh -i new-project.pem ec2-user@3.210.205.42
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Feb 18 05:05:26 2026 from 10.0.1.24
[ec2-user@ip-10-0-1-24 ~]$ ssh -i new-project.pem ec2-user@10.0.3.205
The authenticity of host '10.0.3.205 (10.0.3.205)' can't be established.
ED25519 key fingerprint is SHA256:81VNNl7tSQ9Iyn/Swb6C2gFPLFhovO+kvScgSq7GjQs.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.3.205' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-3-205 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-02-18 05:16:17 UTC; 7min ago
       Docs: man:httpd.service(8)
   Main PID: 3854 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1120)
     Memory: 12.9M
        CPU: 310ms
     CGroup: /system.slice/httpd.service
             ├─3854 /usr/sbin/httpd -DFOREGROUND
             ├─3943 /usr/sbin/httpd -DFOREGROUND
             ├─3948 /usr/sbin/httpd -DFOREGROUND
             ├─3949 /usr/sbin/httpd -DFOREGROUND
             └─3950 /usr/sbin/httpd -DFOREGROUND

Feb 18 05:16:17 ip-10-0-3-205.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Feb 18 05:16:17 ip-10-0-3-205.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Feb 18 05:16:17 ip-10-0-3-205.ec2.internal httpd[3854]: Server configured, listening on: port 80
[ec2-user@ip-10-0-3-205 ~]$ curl localhost
<h1>Healthy Application</h1>
[ec2-user@ip-10-0-3-205 ~]$
```

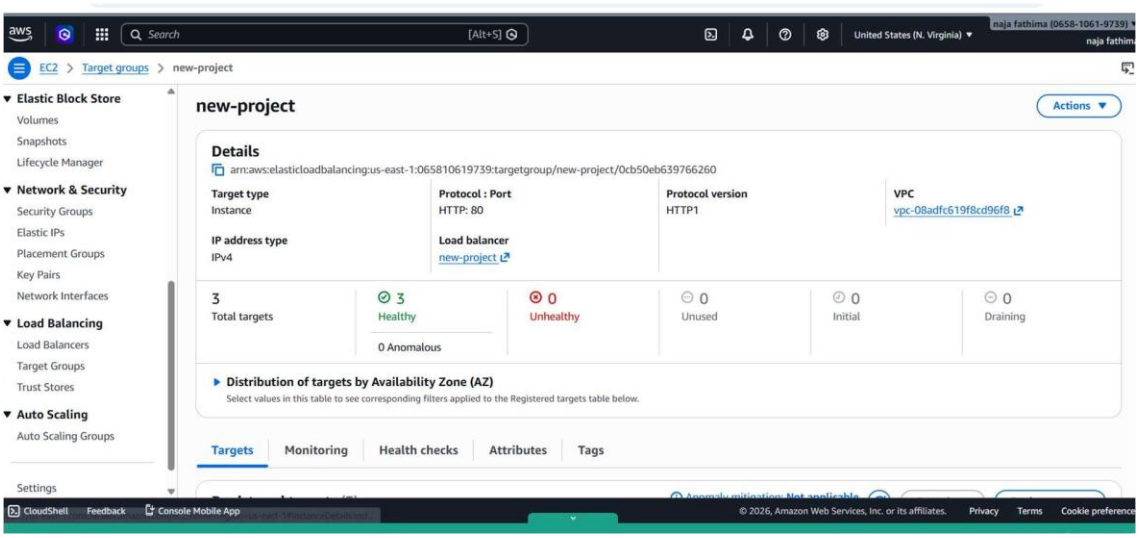
Step 3: Created Auto scaling group

- Created auto scaling group named newproject using new-project launch template.
  Set desired capacity=2 , minimum=1 and maximum=3
- Attached it to the target group.

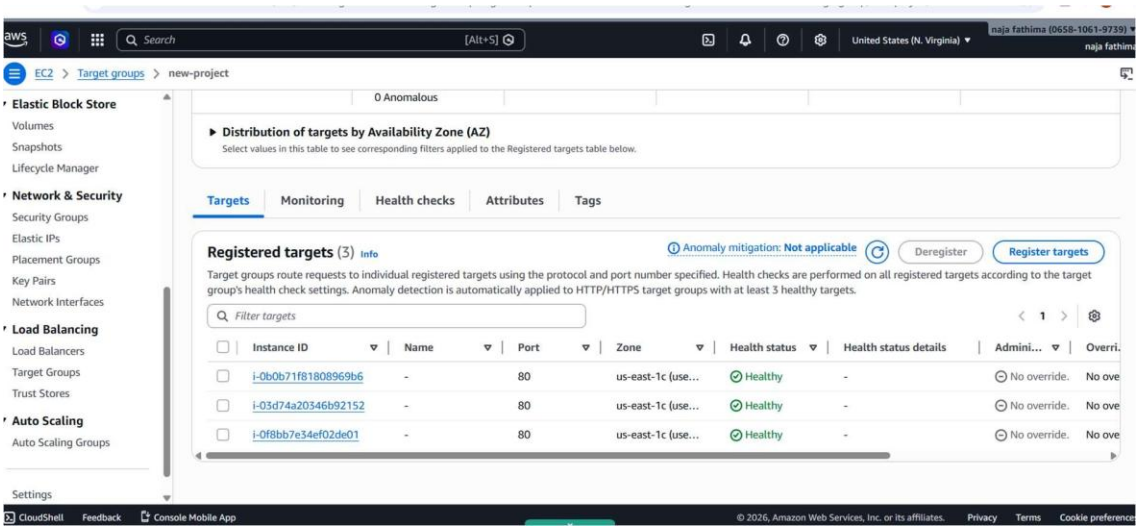


Step 4:Created an Application Load Balancer and forwarded to target group.

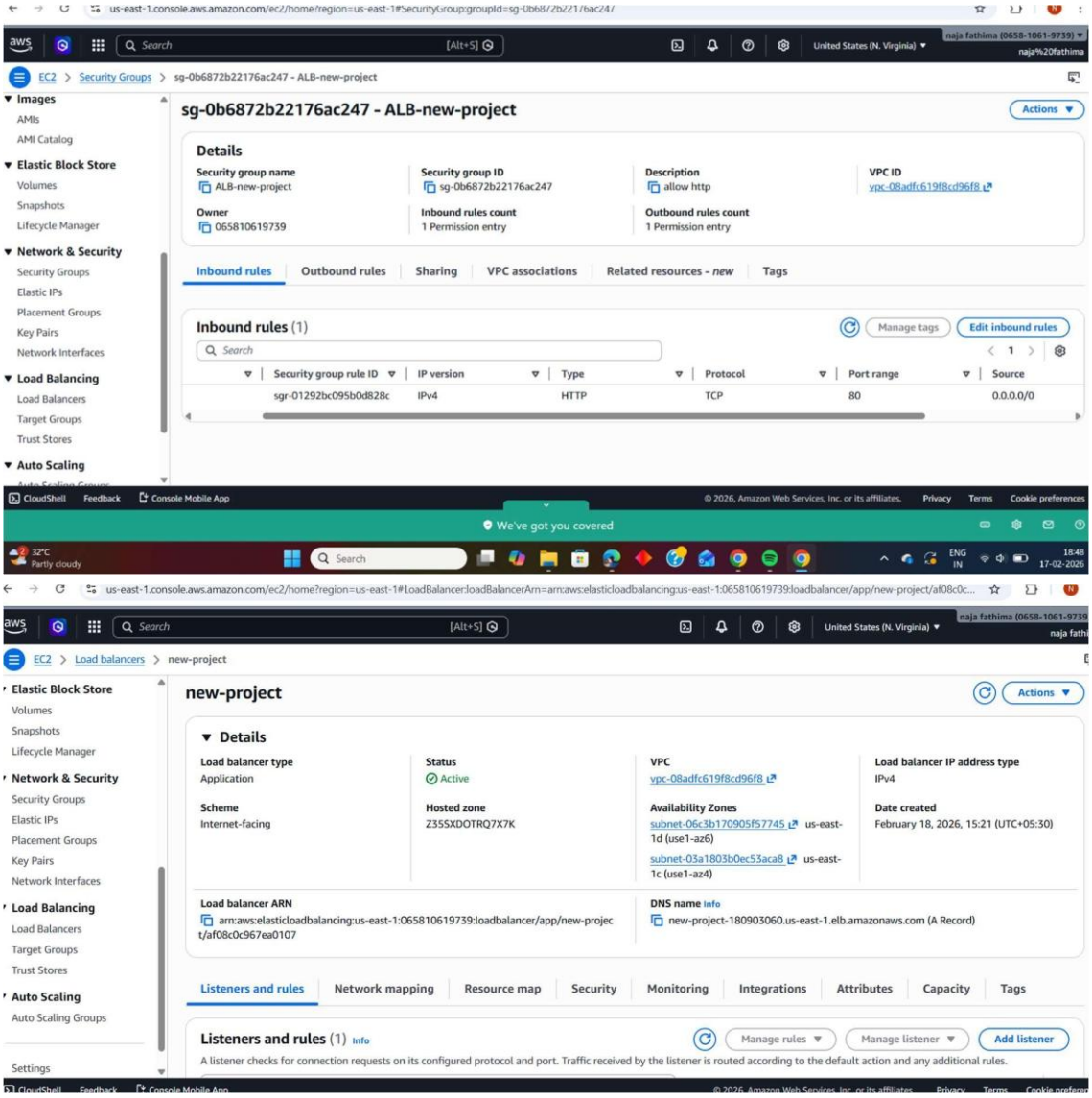
- Created target group named new-project
- Target group allow HTTP protocol.
- Target group is created in new-private-1 subnet.



- Verified the health status of instances. □
  Health status = Healthy.

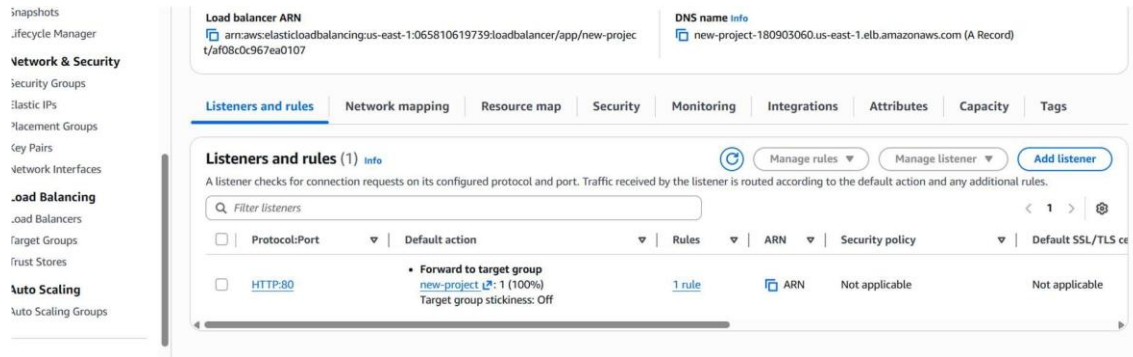


- Created Applica on Load balancer named new-project in two availability Zone.
- Placed the ALB in public subnet.
  ALB is exposed to the Internet.
- The security group of ALB allows HTTP request from internet.



- Added listener rule

  The request from HTTP (port 80) is forwarded to the new-project target group.



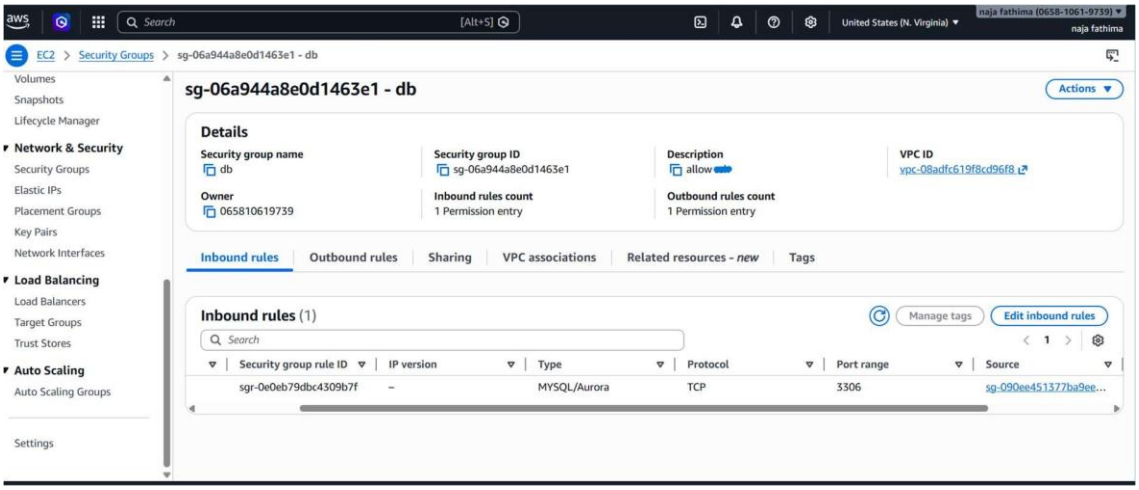
- Verified that the applica on can be accessed using ALB DNS name

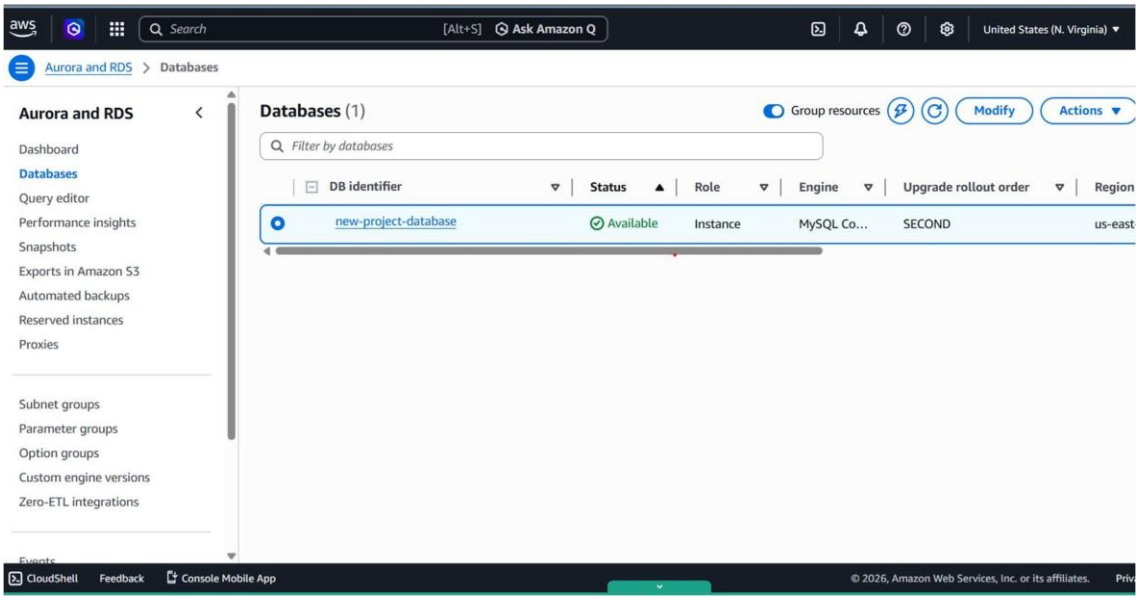
Healthy Application

Step 5:Created a MySQL RDS

- Created MySQL RDS in new-private-2 subnet.
- Database is placed in separate subnet for strong security.

  Created a database security group which only allow EC2 instance to access the database.
- Only EC2 can communicate with database to read/write data.

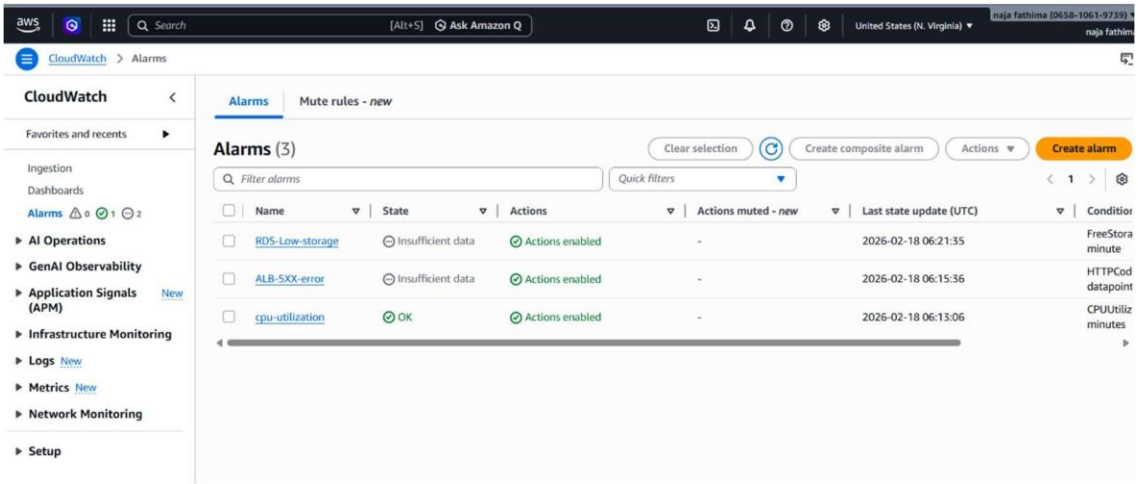




Step 6: Created CloudWatch monitoring

- Created CloudWatch alarm for CPU u liza on >70%, ALB 5XX errors, RDS low storage.

CloudWatch alarm:



Step 7: Created Custom metrics.

- Installed CloudWatch Agent to monitor memory usage and disk usage.

```
ec2-user@ip-10-0-3-205:~
[ec2-user@ip-10-0-3-205 ~]$ sudo dnf install amazon-cloudwatch-agent -y
Last metadata expiration check: 1:34:21 ago on Wed Feb 18 05:16:11 2026.
Dependencies resolved.
================================================================================
 Package                  Arch     Version                  Repository     Size
================================================================================
Installing:
 amazon-cloudwatch-agent  x86_64   1.300062.1-1.amzn2023   amazonlinux    67 M

Transaction Summary
================================================================================
Install  1 Package

Total download size: 67 M
Installed size: 228 M
Downloading Packages:
amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x  66 MB/s |  67 MB     00:01
--------------------------------------------------------------------------------
Total                                            63 MB/s |  67 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x86_64   1/1
create group cwagent, result: 0
create user cwagent, result: 0

  Installing       : amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x86_64   1/1
  Running scriptlet: amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x86_64   1/1
  Verifying        : amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x86_64   1/1

Installed:
  amazon-cloudwatch-agent-1.300062.1-1.amzn2023.x86_64

Complete!
[ec2-user@ip-10-0-3-205 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-clo
udwatch-agent-config-wizard
Starting config-wizard, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with argu
ments: [config-wizard]============================================================
======
= Welcome to the Amazon CloudWatch Agent Configuration Manager =
=                                                              =
= CloudWatch Agent allows you to collect metrics and logs from =
= your host and send them to CloudWatch. Additional CloudWatch =
= charges may apply.                                           =
================================================================
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
I1 imds retry client will retry 1 timesAre you using EC2 or On-Premises hosts?
```

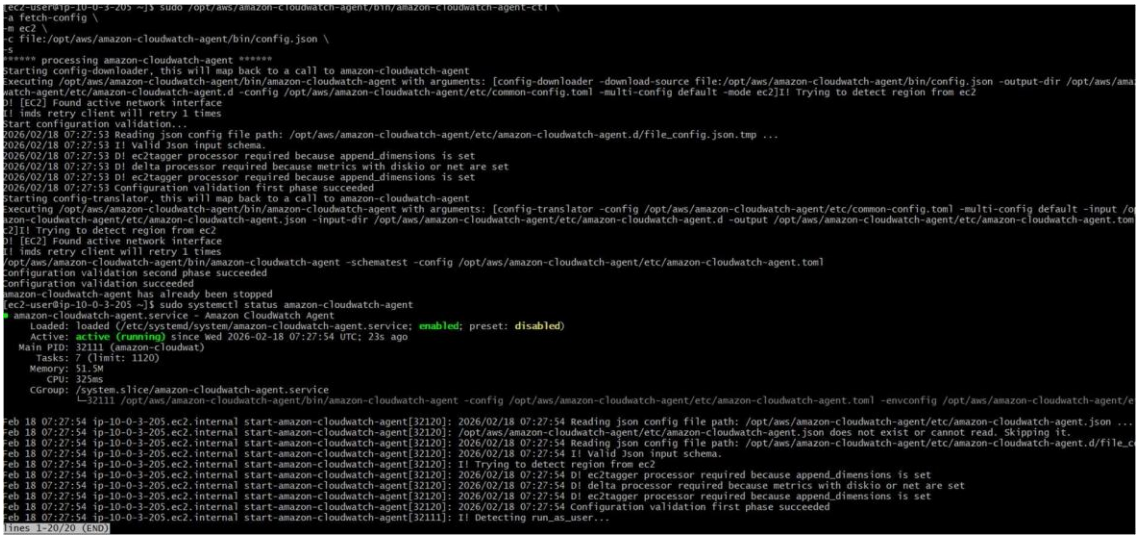
```
================================================================
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
I! imds retry client will retry 1 timesAre you using EC2 or On-Premises hosts?
1. EC2
2. On-Premises
default choice: [1]:
1
Which user are you planning to run the agent?
1. cwagent
2. root
3. others
default choice: [1]:
2
Do you want to turn on StatsD daemon?
1. yes
2. no
default choice: [1]:
2
Do you want to monitor metrics from CollectD? WARNING: CollectD must be installe
d or the Agent will fail to start
1. yes
2. no
default choice: [1]:
2
Do you want to monitor any host metrics? e.g. CPU, memory, etc.
1. yes
2. no
default choice: [1]:
1
Do you want to monitor cpu metrics per core?
1. yes
2. no
default choice: [1]:
2
Do you want to add ec2 dimensions (ImageId, InstanceId, InstanceType, AutoScalin
gGroupName) into all of your metrics if the info is available?
1. yes
2. no
default choice: [1]:
2
Do you want to aggregate ec2 dimensions (InstanceId)?
1. yes
2. no
default choice: [1]:
2
Would you like to collect your metrics at high resolution (sub-minute resolution
)? This enables sub-minute resolution for all metrics, but you can customize for
 specific metrics in the output json file.
1. 1s
2. 10s
3. 30s
4. 60s
```

```
1
Do you have any existing CloudWatch Log Agent (http://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/AgentReference.html) configuration file to import for migration?
1. yes
2. no
default choice: [2]:
1
What is the file path for the existing cloudwatch log agent configuration file?
default choice: [/var/awslogs/etc/awslogs.conf]
/var/log/httpd/access_log
Do you want to monitor any log files?
1. yes
2. no
default choice: [1]:
1
Log file path:
/var/log/httpd/error_log
Log group name:
default choice: [error_log]

Log group class:
1. STANDARD
2. INFREQUENT_ACCESS
default choice: [1]:
1
Log stream name:
default choice: [{instance_id}]

Log Group Retention in days
1. -1
2. 1
3. 3
4. 5
5. 7
6. 14
7. 30
8. 60
9. 90
10. 120
11. 150
12. 180
13. 365
14. 400
15. 545
16. 731
17. 1096
18. 1827
19. 2192
20. 2557
21. 2922
22. 3288
23. 3653
default choice: [1]:
5
Do you want to specify any additional log files to monitor?
1. yes
2. no
default choice: [1]:
```

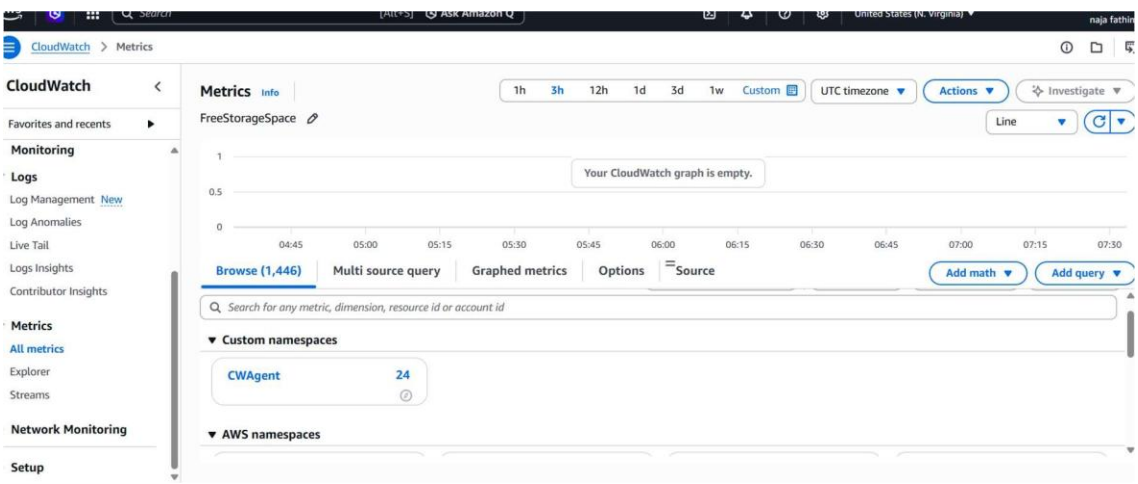
ec2-user@ip-10-0-3-205:~

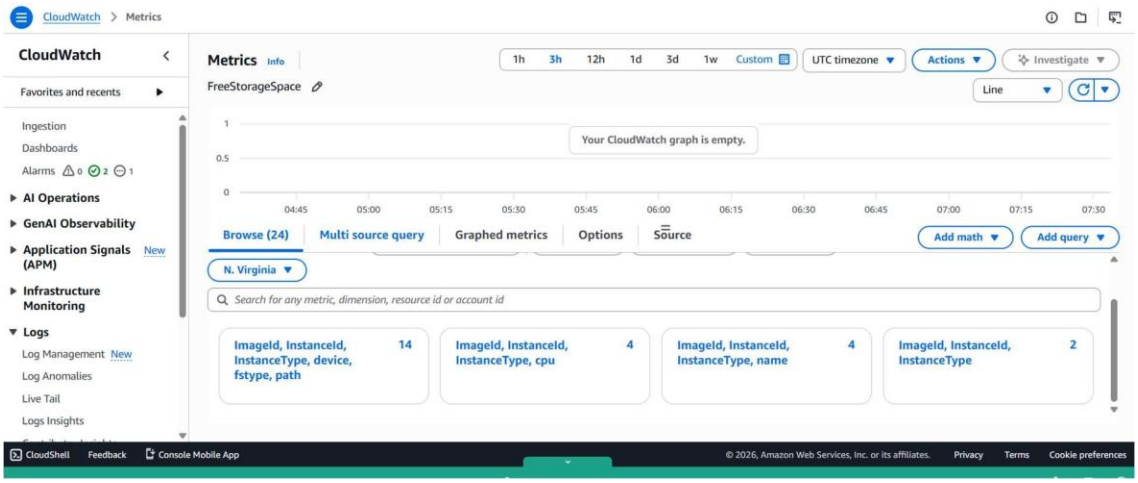
```
default choice: [1]:
2
Existing config JSON identified and copied to:  /opt/aws/amazon-cloudwatch-agent/etc/backup-configs
Saved config file to /opt/aws/amazon-cloudwatch-agent/bin/config.json successfully.
Current config as follows:
{
        "agent": {
                "metrics_collection_interval": 60,
                "run_as_user": "root"
        },
        "logs": {
                "logs_collected": {
                        "files": {
                                "collect_list": [
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
                                        },
                                        {
                                                "file_path": "",
                                                "log_group_name": "",
                                                "retention_in_days": -1
```

```
he config file is also located at /opt/aws/amazon-cloudwatch-agent/bin/config.json.
dit it manually if needed.
o you want to store the config in the SSM parameter store?
. yes
. no
efault choice: [1]:

rogram exits now.
ec2-user@ip-10-0-3-205 ~]$ sudo systemctl start amazon-cloudwatch-agent
ec2-user@ip-10-0-3-205 ~]$ sudo systemctl status amazon-cloudwatch-agent
 amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
    Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; disabled; preset: disabled)
    Active: activating (auto-restart) (Result: exit-code) since Wed 2026-02-18 07:14:54 UTC; 29s ago
   Process: 31253 ExecStart=/opt/aws/amazon-cloudwatch-agent/bin/start-amazon-cloudwatch-agent (code=exited, status=1/FAILURE)
  Main PID: 31253 (code=exited, status=1/FAILURE)
       CPU: 85ms
ec2-user@ip-10-0-3-205 ~]$ sudo systemctl enable amazon-cloudwatch-agent
reated symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/systemd/system/amazon-cloudwatch-agent.service.
ec2-user@ip-10-0-3-205 ~]$ sudo systemctl status amazon-cloudwatch-agent
 amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
    Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
    Active: activating (auto-restart) (Result: exit-code) since Wed 2026-02-18 07:15:55 UTC; 14s ago
  Main PID: 31304 (code=exited, status=1/FAILURE)
       CPU: 82ms
ec2-user@ip-10-0-3-205 ~]$ |
```

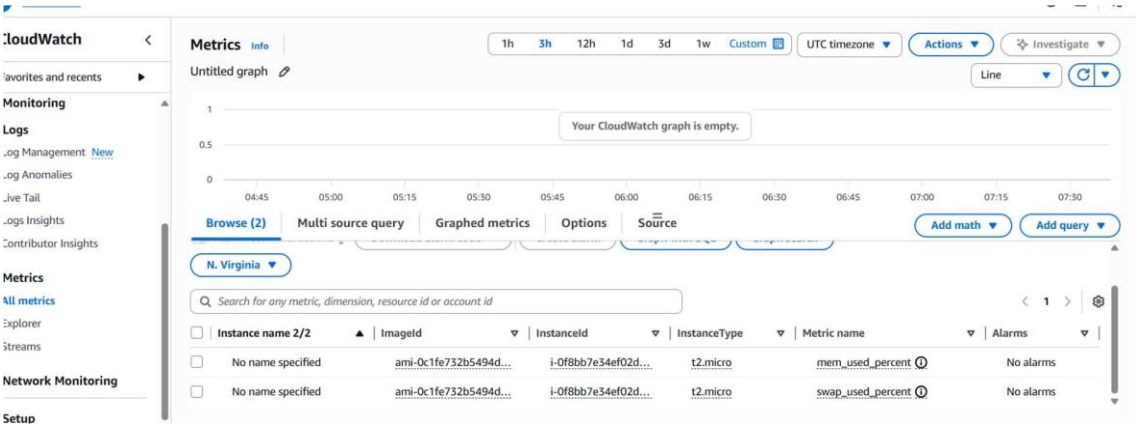


- Verified that the custom metrics is published under the dedicated namespace.

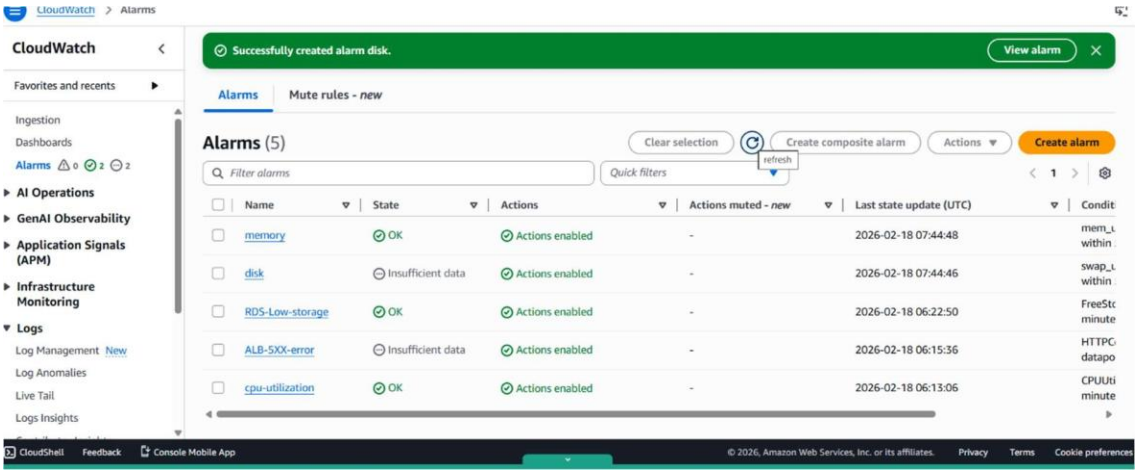




- Verified that the custom metrics are shown

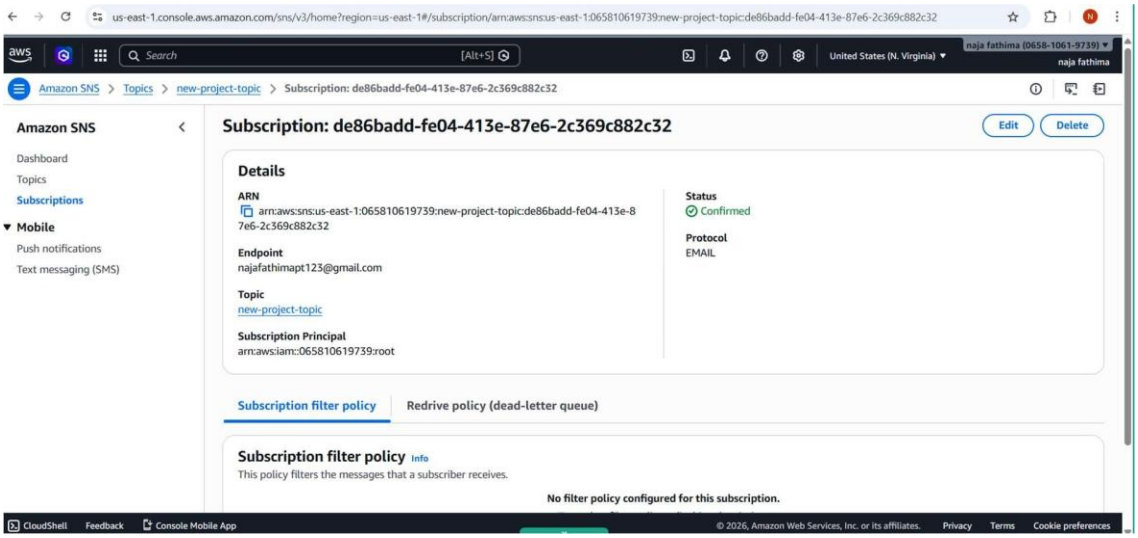


- Created CloudWatch alarms named memory and disk for custom metrics.



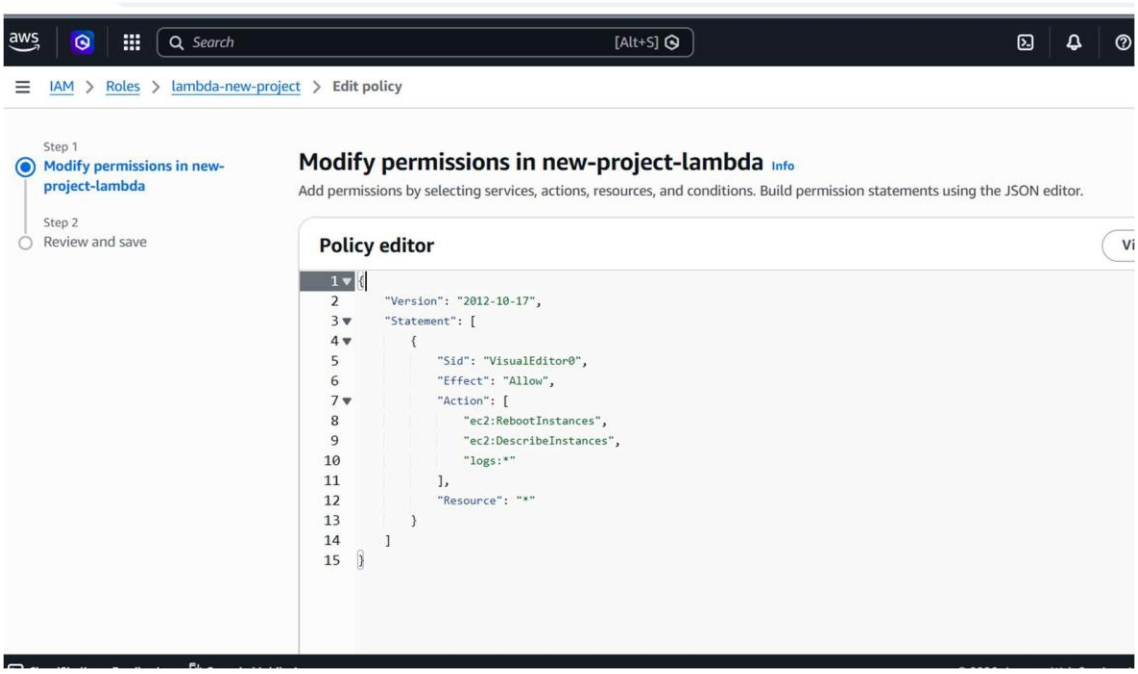
Step 8: Created SNS

- Created SNS topic and added email as subscrip on.
- A ached SNS to CloudWatch alarms
- SNS will send alerts via email when any of the CloudWatch metrics crosses the threshold value.

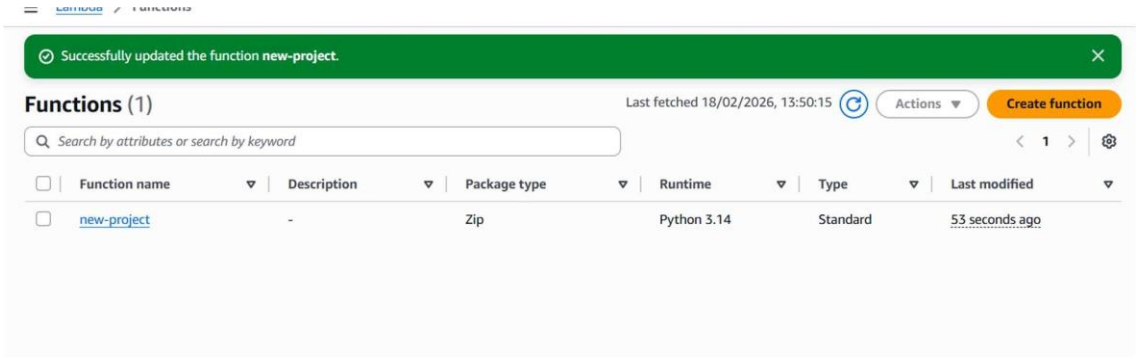


Step 9:Created Lambda function

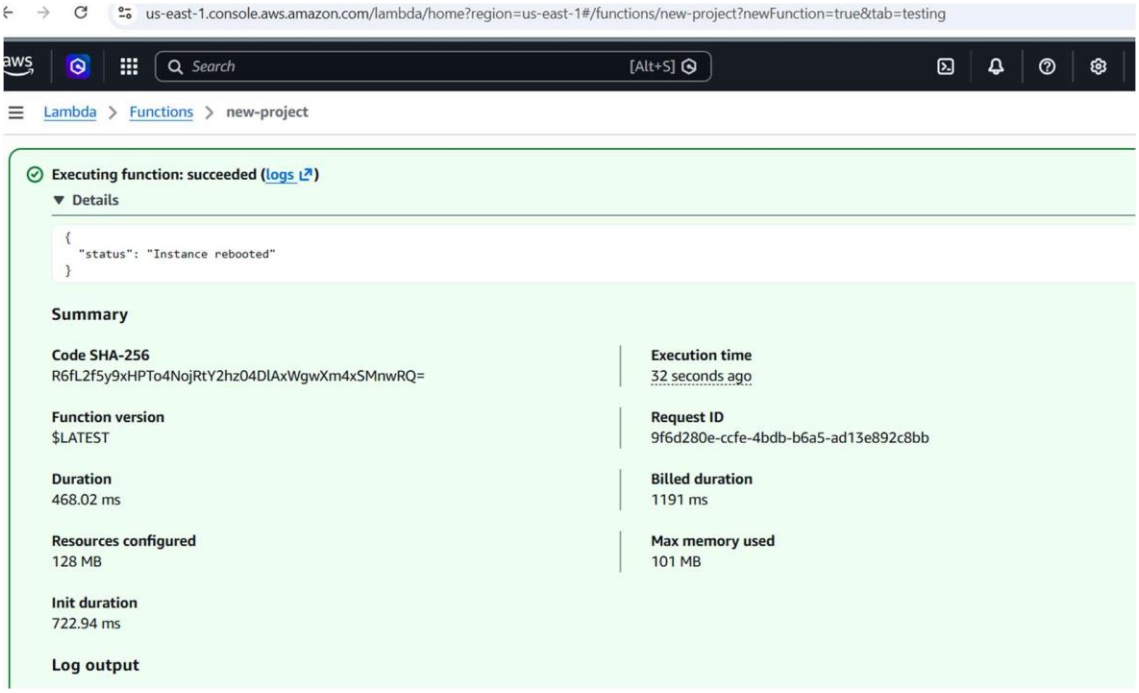
- Created an IAM role with ec2:DescribeInstance, ec2:RebootInstance and logs:* permissions for Lambda



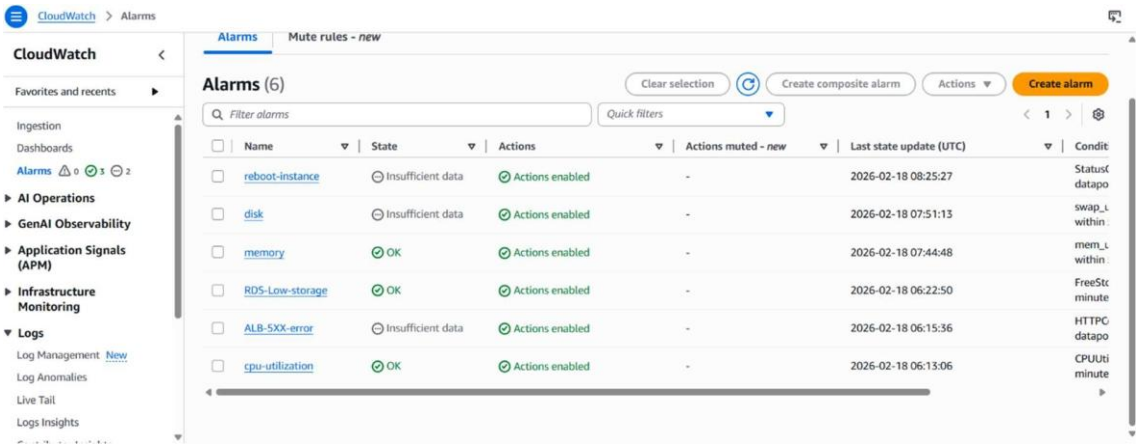
- Created a lambda func on named newproject and a ached the created IAM role to perform specified ac on on EC2(To reboot unhealthy instance)



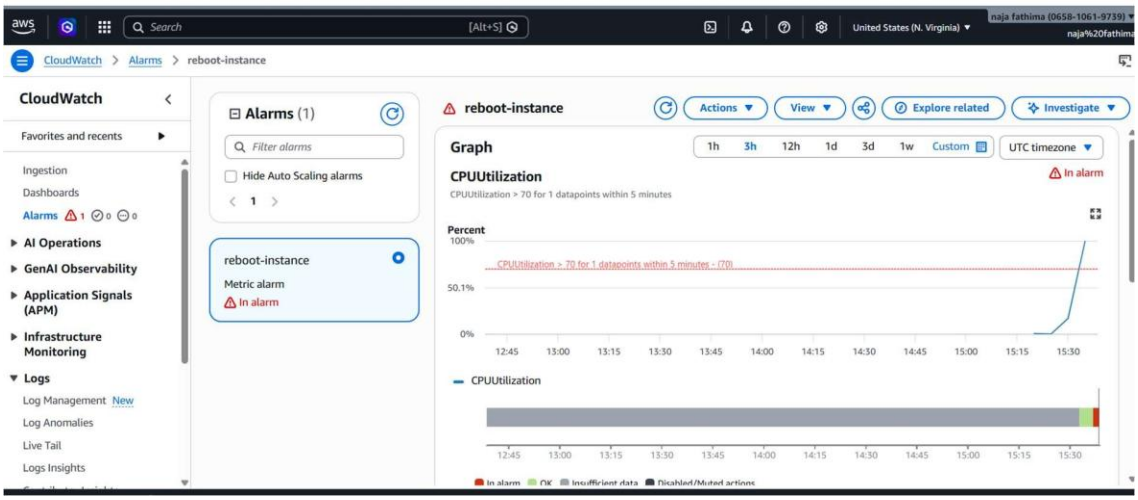
- Deployed the code and tested with sample input.



- Created CloudWatch alarm named rebootinstance to trigger Lambda



- Tested it



- Lambda executes and EC2 reboots automatically

Output :

The application is accessed using ALB DNS name

Healthy Application